Mohammad Imran Hossain

MSCA Doctoral Candidate | Medical AI & Cancer Research

Stockholm, Sweden | hossainimran.maia@gmail.com | github.com/imran-maia |
linkedin.com/in/mohammad-imran-hossain-23721812a

Doctoral researcher with training across computational pathology, multimodal AI, and medical image analysis. Current doctoral work at Karolinska Institutet focuses on AI-driven breast cancer risk prediction and personalized screening using mammography and breast MRI. Member of the EU-funded HER-CARE MSCA Doctoral Network (DC09).

Research Interests

Computational pathology; multimodal AI for healthcare; medical image analysis; spatial transcriptomics; weakly supervised learning; self-supervised learning; segmentation; registration; image reconstruction; AI-assisted diagnosis.

Research Experience

MSCA Doctoral Candidate - HER-CARE (DC09)

Karolinska Institutet, Stockholm, Sweden | 2026 - Present

Department of Medical Epidemiology and Biostatistics. AI-driven breast cancer risk prediction and personalized screening using mammography and breast MRI. EU-funded MSCA Doctoral Network.

Research Engineer - Bioinformatics & AI

Institut Curie - PSL Research University, Paris, France | From Oct 2025 (prior role)

• Developed multimodal computational pipelines integrating histopathology and spatial transcriptomics to study tumor microenvironment dynamics.

• Contributed to clinically grounded AI workflows for cancer biology with a focus on reproducible analysis and manuscript preparation.

Research Intern - Bioinformatics & Spatial Transcriptomics

Institut Curie / Sanofi collaboration, Paris, France | Feb 2025 - Aug 2025

• Built analysis pipelines for Visium V2 and Visium HD spatial transcriptomics data.

• Identified eight candidate genes potentially mediating CAF-driven T-cell exclusion in lung squamous cell carcinoma.

• Contributed to ongoing manuscript preparation.

Research Intern - Computational Pathology

National Center for Scientific Research (CNRS), France | Feb 2024 - Jul 2024

• Benchmarked fully supervised, weakly supervised, and self-supervised methods for homologous recombination deficiency detection in breast and ovarian cancer whole-slide images.

• Evaluated attention-based multiple instance learning and foundation-model-based pipelines.

• Best performance reached AUC 0.78 on breast cancer and 0.68 on ovarian cancer.

Visiting Researcher - Medical Image Analysis

Diagnostic Image Analysis Group, Radboud University Medical Center, Netherlands | Aug 2023 - Oct 2023

• Developed preprocessing pipelines for k-space undersampling and evaluated deep learning models for real-time MRI reconstruction in interventional radiology.

Education

Master of Computer Science, Bioinformatics and Modeling

Sorbonne University, France | 2024 - 2025

Grade: 14.85/20 | Rank: 1st of 8 | Best Thesis Award
Thesis: Identification of Tumor Gene Signatures Underlying Fibroblast-Mediated T Cell Exclusion in Lung Cancer Using Spatial Transcriptomics.

Erasmus Mundus Joint Master's in Medical Imaging and Applications
University of Girona, University of Cassino, University of Burgundy | 2022 - 2024
Grade: 8.30/10 | Fully funded by the European Union (EUR 42,000).

17th EXCITE Summer School on Biomedical Imaging
ETH Zurich and University of Zurich, Switzerland | 2023
4 ECTS.

B.Sc. in Electrical and Electronic Engineering
United International University, Bangladesh | 2018 - 2022
GPA: 3.97/4.00 | Summa Cum Laude | Rank: 1st of 120.

Preprints and Manuscripts

Tumor Gene Signatures Underlying Fibroblast-Mediated T Cell Exclusion in Lung Cancer
Institut Curie / Sanofi collaboration | In preparation
Hossain M.I. et al. Manuscript in preparation, 2026.

Comparative Study of Probabilistic Atlas and Deep Learning for Brain Tissue Segmentation
ArXiv 2024
Hossain M.I., Amin M.Z., et al. arXiv:2411.05456

Deep Learning and Classical Computer Vision in Medical Image Analysis
ArXiv 2025
Tweneboah A.D., Hossain M.I. (co-author), et al. arXiv:2502.19258

Selected Projects

Spatial Transcriptomics Analysis of Tumor Immune Exclusion
Institut Curie / Sanofi | 2025
Visium HD pipeline for identifying CAF-mediated T-cell exclusion signatures in lung squamous cell carcinoma.

HRD Detection in Cancer WSIs via Foundation Models and MIL
CNRS | 2024
Benchmarked AB-MIL, CLAM, Trans-MIL, and foundation models for homologous recombination deficiency detection.

Cardiac Structure Segmentation from 2D Echocardiograms
Sorbonne University | 2025
nnU-Net pipeline for echocardiogram segmentation. Dice scores: LV endocardium 0.94, LV epicardium 0.91, left atrium 0.93.

Breast Cancer Subtype Classification via Multi-omics Integration
Sorbonne University | 2024
XGBoost pipeline integrating DNA methylation, CNV, mRNA, and miRNA data for subtype classification. Balanced multiclass accuracy: 0.90.

Awards and Distinctions

- Best Thesis Award, Sorbonne University.
- Ranked 2nd in the MAIA Brain Tissue Segmentation Challenge, University of Girona.
- Erasmus Mundus Scholarship awarded by the European Union.
- Summa Cum Laude, United International University.